Running the sms-ecoli ↔ vEcoli comparison harness

A one-command, fully-local way to compare the sms-ecoli whole-cell model against a Covert-Lab vEcoli fork — and to transfer a process from the fork into sms-ecoli and compare that too.

1 · What this compares, and why

Two whole-cell *E. coli* models are run on the same nutrient conditions and compared at matched cell-cycle timepoints. One HTML report is produced.

Role	Repository	What it is
Candidate (measured)	`sms-ecoli`	The whole cell reimplemented on the **process-bigraph** engine (the *v2ecoli* lineage). The harness lives here.
Reference (baseline)	`vEcoli-private`	The Covert-Lab **vEcoli** whole-cell model (a private fork), run natively.

Beyond a plain comparison, a fork *config* can **swap in a process** (e.g. MetabolismRedux). That process is auto-converted from vEcoli (v1) to process-bigraph (v2) and injected into the sms-ecoli candidate — so you compare the *same* process running in both engines. The report embeds the full source of every transferred process.

2 · What to download

Clone both private repos side by side (e.g. under `~/code/`). Access is via the `CovertLabEcoli` GitHub org.

```
git clone https://github.com/CovertLabEcoli/sms-ecoli.git       ~/code/sms-ecoli
git clone https://github.com/CovertLabEcoli/vEcoli-private.git  ~/code/vEcoli-private
```

Repository	Purpose
`CovertLabEcoli/sms-ecoli`	The comparison harness + the candidate model.
`CovertLabEcoli/vEcoli-private`	The reference fork (any vEcoli checkout works).

3 · One-time setup

Requires `uv` (Python package manager) and a C compiler (Xcode command-line tools on macOS). Build the sms-ecoli environment:

```
cd ~/code/sms-ecoli
uv sync            # creates .venv with all dependencies (pinned)
```

No network access or external service is needed at run time. The vEcoli fork does **not** need its own environment — the harness loads its process code directly.

4 · Run it — one command

```
cd ~/code/sms-ecoli
.venv/bin/python scripts/run_local_comparison.py \
    --fork ~/code/vEcoli-private \
    --configs basal configs/metabolism_redux.json \
    --out out/comparison --open
```

Flag	Meaning
`--fork`	Path to the vEcoli fork to compare against (the reference).
`--configs`	One or more items: a **condition name** (`basal`, `with_aa`, `succinate`, `no_oxygen`, `acetate`) and/or a path to a **fork config JSON** (e.g. `configs/metabolism_redux.json`).
`--out`	Output directory (default `out/comparison`).
`--open`	Open the finished report in your browser.
`--steps` / `--seeds`	Optional: sim length per engine (default 300) / seeds per config (default 1).

> **First run builds two caches automatically.** The sms-ecoli ParCa cache (fast, from a shipped fixture) and the fork's ParCa cache (**heavy — about 10 minutes**, built once and reused). Later runs skip both.

5 · What you get

A single self-contained HTML report (`out/comparison/standardized_comparison_report.html`):

- **Repositories compared** — the two repos with their branch + commit, so the run is reproducible.
- **Per-condition comparison** — matched gen-1 % difference for cell / dry / protein / RNA mass and growth rate (candidate vs reference), with a within-tolerance / drift / mismatch verdict.
- **Transferred process source** — for any config that swaps in a process, the converted process is listed and its **full source code** (read from the fork) is embedded.

6 · Choosing what to compare

- **Plain conditions** — pass condition names to compare the standard cell in different media: `--configs basal with_aa succinate no_oxygen acetate`.
- **Process transfers** — pass a fork config that swaps or adds a process. Any config under `vEcoli-private/configs/` that sets `swap_processes` (e.g. `metabolism_redux.json`) transfers that process into sms-ecoli. Raw fork configs work as-is; needed unit bridging is applied automatically.

Everything runs locally on one machine. No agents, no cloud, no external services — just the two repositories and one command.